

Peptidoglycan-tethered and free forms of the Braun lipoprotein are in dynamic equilibrium in *Escherichia coli*

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Abstract

Peptidoglycan (PG) is a giant macromolecule that completely surrounds bacterial cells and prevents lysis in hypo-osmotic environments. This net-like macromolecule is made of glycan strands linked to each other by two types of transpeptidases that form either 4 → 3 (PBPs) or 3 → 3 (LDTs) cross-links. Previously, we devised a heavy isotope-based PG full labeling method coupled to mass spectrometry to determine the mode of insertion of new-subunits into the expanding PG network (Atze et al., 2022). We showed that PG polymerization operates according to different modes for the formation of the septum and of the lateral cell walls, as well as for bacterial growth in the presence or absence of β-lactams in engineered strains that can exclusively rely on LDTs for PG cross-linking when drugs are present. Here, we apply our method to the resolution of the kinetics of the reactions leading to the covalent tethering of the Braun lipoprotein (Lpp) to PG and the subsequent hydrolysis of that same covalent link. We find that Lpp and disaccharide-peptide subunits are independently incorporated into the expanding lateral cell walls. Newly synthesized septum PG appears to contain small amounts of tethered Lpp. LDTs did mediate intense shuffling of Lpp between PG stems leading to a dynamic equilibrium between the PG-tethered and free forms of Lpp.

eLife assessment

This **useful** study describes a single set of label-chase mass spectrometry experiments to confirm the molecular function of YafK as a peptidoglycan hydrolase, and to describe the timing of its attachment to the peptidoglycan. Confirmation of the molecular function of YafK will be helpful in further studies to examine the function and regulation of the outer membrane-peptidoglycan link in bacteria. The evidence supporting the molecular function of YafK and that lpp molecules are shuffled on and off the peptidoglycan is **solid**, however, data supporting conclusions relating to the locations of lpp-peptidoglycan attachment are **incomplete**. The work will be of interest to researchers studying lipoproteins in gram negative bacteria.

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The PG is an essential component of the bacterial cell envelope that counteracts the turgor pressure of the cytoplasm thereby preventing the cell from bursting and lysing in hypoosmotic environments (**Fig. 1A**) (Rojas et al., 2018). Structural studies of the PG (**Fig. 1B**) revealed that this giant macromolecule consists of glycan strands linked to each other by transpeptidases belonging to the PBP and LDT families, which form 4 → 3 and 3 → 3 cross-links, respectively (Glauner et al., 1988). PBPs and LDTs contain different catalytic nucleophiles (Ser *versus* Cys) and use different stem peptides as the acyl donor substrate (pentapeptide *versus* tetrapeptide) (**Fig. 1C**) (Mainardi et al., 2005). LDTs forming 3 → 3 cross-links are unessential enzymes (Magnet et al., 2008), which can participate in PG maintenance (Morè et al., 2019) and in β-lactam resistance acquired *in vitro* (Voedts et al., 2022). The LDT family includes additional enzymes (**Fig. 1C**) that catalyze either the tethering of Lpp to PG (ErfK, Ybis, and YcfS) (Magnet 2007) or the hydrolysis of the resulting covalent link (YafK) (Bahadur et al., 2021); (Winkle et al., 2021). Lpp is the most abundant lipoprotein (10⁶ copies per cell), about one third of which is attached to PG. This dense connection between the PG and the outer membrane (OM) is thought to stabilize the bacterial envelope because mutants deficient in the tethering of Lpp to PG are viable but display increased vesiculation and lethal susceptibility to EDTA (Sanders and Pavelka, 2013). In addition, Lpp controls the distance between the inner (IM) and outer membranes, a structural feature that is critical for sensing stresses that affect the bacterial envelope (Asmar et al., 2017); (Mathelié-Guinlet et al., 2020).

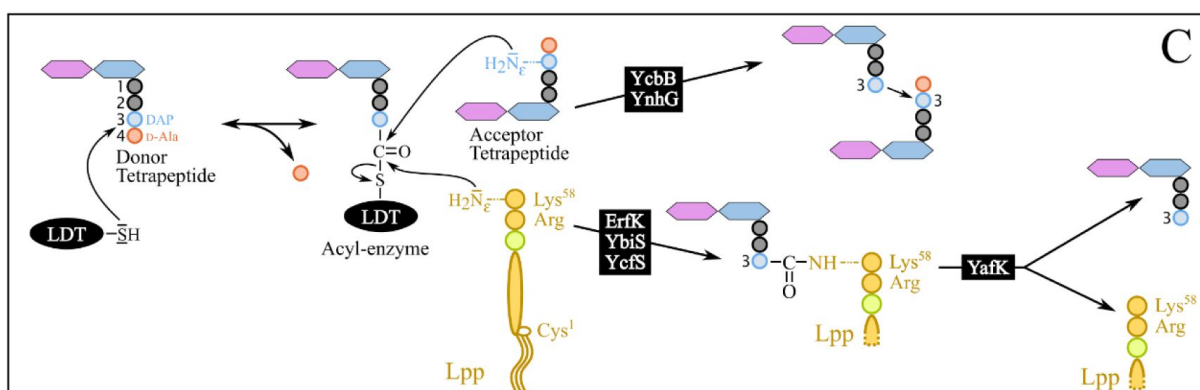
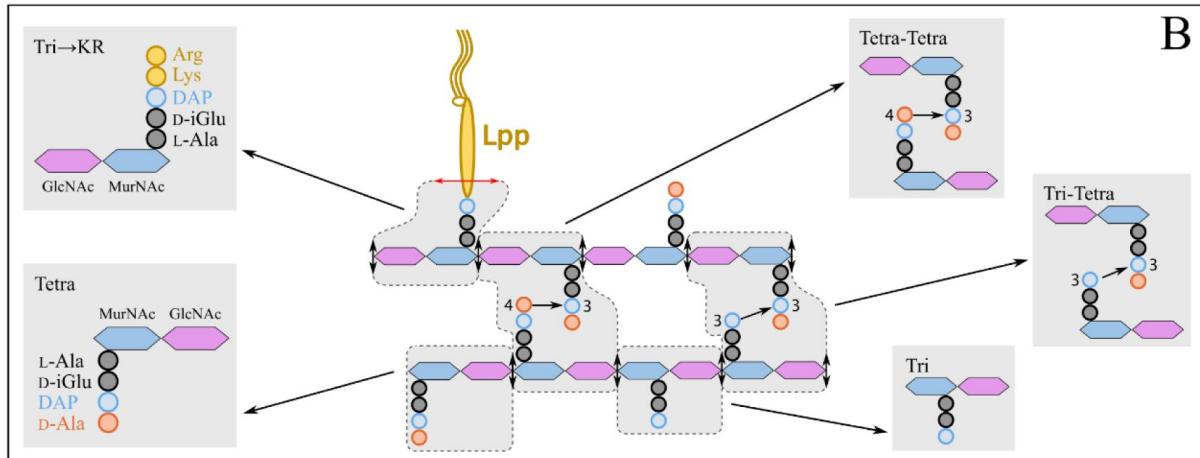
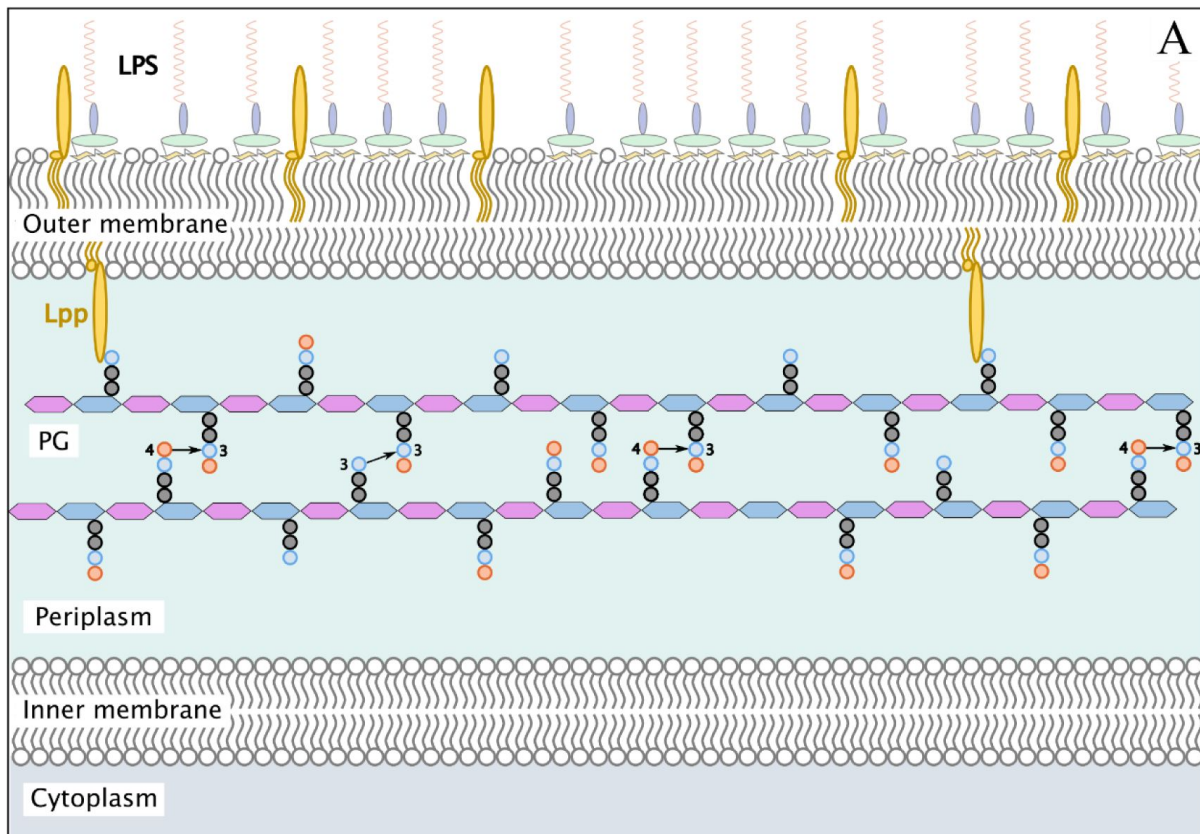


Figure 1.

Structure and synthesis of the cell envelope of *Escherichia coli*.

(A) Structure of the main polymers of the envelope. The envelope of Gram-negative bacteria contains two membranes. The inner membrane (IM) is made of phospholipids. The outer membrane (OM) is asymmetric, with phospholipids in the inner leaflet and lipopolysaccharides (LPS) in the outer leaflet. The PG is located in the periplasmic space delimited by the IM and the OM. The PG consists of glycan chains linked to each other by short peptides. The glycan chains consist of alternating $\beta 1 \rightarrow 4$ -linked *N*-acetyl-glucosamine (GlcNAc) and *N*-acetyl muramic acid (MurNAc) residues. The sequence of the stem peptide linked to the D-lactoyl group of MurNAc, as assembled in the cytoplasm, is L-Ala¹-D-iGlu²-DAP³-D-Ala⁴-D-Ala⁵, in which DAP is a diaminopimelyl residue. Polymerization of this disaccharide-pentapeptide subunit by glycosyltransferases results in linear glycan strands that are inserted into the expanding PG network by the combined actions of space-making hydrolases and transpeptidases. The majority of the cross-links connecting D-Ala⁴ of a donor stem peptide to DAP³ of an acceptor stem peptide ($4 \rightarrow 3$ cross-links) is made by D,D-transpeptidases belonging to the penicillin-binding protein (PBP) family that are the essential targets of β -lactam antibiotics. L,D-transpeptidases (LDTs) catalyze the formation of $3 \rightarrow 3$ cross-links connecting two DAP residues (see panel C). The Braun lipoprotein (Lpp) provides a link between the OM and the PG since its C-terminus is covalently bound to DAP residues (panel C), whereas its N-terminal Cys residue carries three alkyl chains, one of which is carried by the amino group of the C-terminal Cys residue and the other two by a glycerol moiety linked to the sulfhydryl of that residue by a thioether bond. These alkyl chains are inserted into the inner leaflet of the OM (PG-tethered form of Lpp) or to the outer leaflet of that membrane (free form of Lpp facing the bacterial cell surface).

(B) Determination of peptidoglycan structure. The sacculi are extracted by the hot SDS procedure, treated with pronase and trypsin (red double arrows), and digested by muramidases that cleave the MurNAc-GlcNAc bonds (black double arrows). The structure of the resulting fragments (enclosed in rounded polygons) is determined by mass spectrometry following reduction of MurNAc residues and separation by *rp*HPLC. Please note that we used the standard nomenclature for transpeptidation products in which the acyl donor and the acyl acceptor appear left and right, respectively, separated by an arrow to indicate the CO-to-NH polarity of the amide bond.

(C) Reactions catalyzed by L,D-transpeptidases. In *E. coli*, the LDT protein family includes six paralogues that catalyze (i) the formation of $3 \rightarrow 3$ cross-links (YcbB and YnhG), (ii) the covalent anchoring of Lpp to PG (Ybis, YcfS, and ErfK), and (iii) the hydrolysis of the resulting PG $\rightarrow$ Lpp bond (YafK). Formation of the $3 \rightarrow 3$ and PG $\rightarrow$ Lpp bonds involves a common catalytic intermediate resulting from the nucleophilic attack of the DAP³-D-Ala⁴ bond of an acyl-donor tetrapeptide stem by the active Cys residue of the LDTs, the release of D-Ala⁴, and the formation of a D-Ala⁴-Cys thioester bond. For the formation of a $3 \rightarrow 3$ cross-link, the side-chain amino group of DAP at the third position of an acyl acceptor stem reacts with the acyl enzyme. For the formation of a PG $\rightarrow$ Lpp bond, the side-chain of the C-terminal Lys residue of Lpp (at position 58) reacts with the acyl enzyme.

The enzymatic activity of YafK was established by incubating PG with purified YafK and detecting the products of the hydrolysis of the PG $\rightarrow$ Lpp bonds (Bahadur et al., 2021 [DOI](#)); (Winkle et al., 2021 [DOI](#)). Intriguingly, the deletion of *yafK* was not associated with any decrease in the overall Lpp-to-PG tethering in the absence of stress (Bahadur et al., 2021 [DOI](#)). The absence of phenotype associated with the deletion of *yafK* implies, in the wild-type strain, either that the enzyme was not produced (or active) or that hydrolysis of PG $\rightarrow$ Lpp bonds by YafK is compensated by the formation of the same number of PG $\rightarrow$ Lpp bonds by LDTs. Here, we used a PG labeling method (Atze et al., 2022) to distinguish between these two possibilities and determine the mode of cross-linking of newly synthesized Lpp to PG.

Results and discussion

Time-resolved characterization of the isotopic composition of the Tri → KR muropeptide by mass spectrometry

To investigate the mode of tethering of Lpp to PG, bacteria were grown in minimal medium containing [^{13}C]glucose and [^{15}N]ammonium chloride, thus affording fully labeled PG. At the late log phase, bacteria were transferred to a light isotope medium and the PG macromolecule was extracted from culture samples collected at various times after the medium switch. Digestion of PG with proteinases and muramidases released soluble disaccharide-peptide fragments (muropeptides) and eliminated the bulk of Lpp except for its C-terminal Arg-Lys dipeptide that remained bound to tripeptide stems (muropeptide Tri → KR in **Fig. 1B**). The isotopic composition of the Tri → KR isotopologues was determined by mass spectrometry (**Figure 2**) and tandem mass spectrometry (Table S1). This analysis revealed that tethering of Lpp to PG did generate Tri → KR isotopologues containing light and heavy moieties at both the donor (tripeptide) and acceptor (Lpp) positions.

Newly synthesized Lpp and stem peptides are independently incorporated into PG

The PG is polymerized from lipid II consisting of the disaccharide-pentapeptide subunit linked to the undecaprenyl lipid carrier by a pyrophosphate bond. In Gram-positive bacteria, proteins are tethered to PG by sortases that use lipid II as the acceptor of the transpeptidation reaction (Drams et al., 2008). Kinetic analysis of the appearance of isotopologues of the Tri → KR muropeptide ruled out this mode of tethering for Lpp. Indeed, this would have resulted in the linking of newly synthesized Lpp to newly synthesized stem peptide in lipid II, thus leading to the formation of Tri → KR isotopologues of the new → new type, which were only detected at late times of the kinetics, when light tetrapeptide stems became abundant in the PG (**Fig. 3A** and **Supplementary Table S1**). The low abundance of Tri → KR isotopologues of the new → new type at early times of the kinetics also indicates that Lpp is not preferentially incorporated at locations in which new disaccharide-peptide subunits are incorporated into the expanding PG. Thus, the tethering of Lpp to PG and the incorporation of newly synthesized subunits are spatially independent.

Tethering of Lpp to septal PG might be limited

The mode of insertion of PG subunits into the expanding PG is different in the side walls and in the septum (de Jonge et al., 1989). One-at-a time insertion of newly synthesized glycan strands into the side-walls leads to a mosaic of new and old subunits gradually enriched in new subunits. In contrast, the septum mainly consists of newly synthesized PG subunits. The low abundance of new → new Tri → KR isotopologues at early time points (**Fig. 3A**) thus suggests that Lpp might be incorporated in very limited amounts into nascent septal PG. **YafK hydrolyzes tripeptide → Lpp bonds.** After the medium switch, the relative abundance of old → old isotopologues is expected to decrease as a result of (i) the synthesis of new Lpp and PG stem peptides leading to a twofold decrease in one generation (*ca.* 60 min) and (ii) the hydrolysis of the tripeptide → Lpp bond by YafK. Indeed, the latter reaction releases heavy Lpp, which may remain unbound (free Lpp) or subsequently cross-linked to newly synthesized peptide stems. The decrease in the relative abundance of old → old isotopologues was slower in the absence of YafK (**Fig. 3B**), thus establishing that YafK is actively hydrolyzing tripeptide → Lpp bonds.

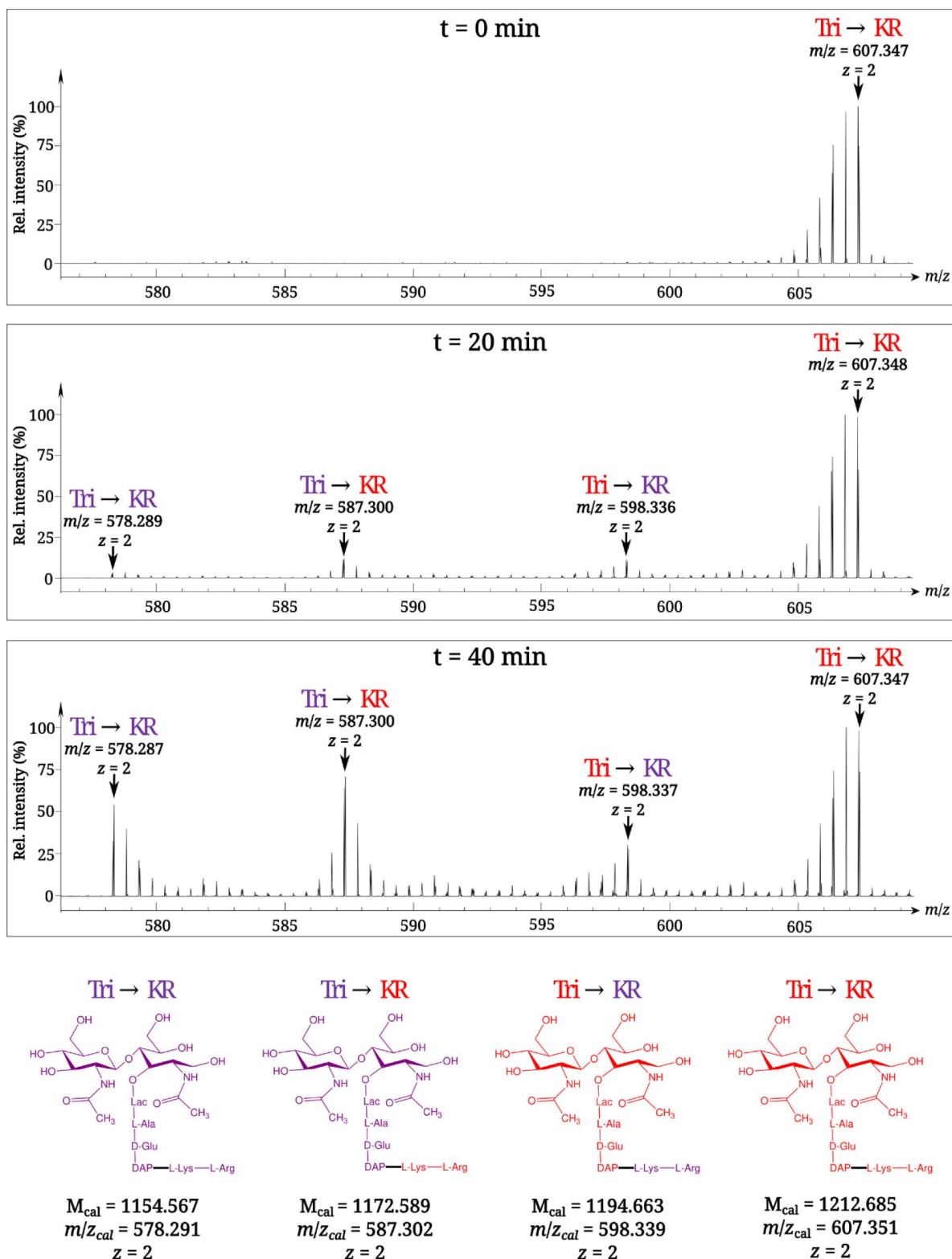


Figure 2.

Time-resolved appearance of differentially labeled Tri → KR mucopeptide isotopologues containing new (light) and old (heavy) Lpp and PG moieties.

Light and heavy residues are indicated in purple and in red, respectively. M_{cal} , calculated monoisotopic mass (see Atze et al., 2022) for the detailed description of the method used to determine M_{cal} values).

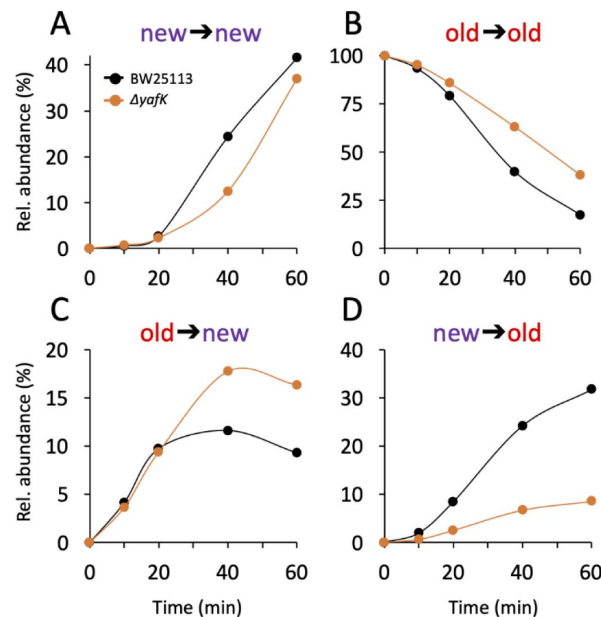


Figure 3.

Kinetics of variations in the relative abundance of Tri → KR isotopologues.

(A) Unlabeled isotopologue containing newly synthesized muropeptide and Lpp moieties (new → new). (B) Fully labeled isotopologue containing existing muropeptide and Lpp moieties (old → old). (C) Hybrid isotopologue containing old and new muropeptide and Lpp moieties, respectively (old → new). (D) Hybrid isotopologue containing new and old muropeptide and Lpp moieties, respectively (new → old). Light and heavy moieties are indicated in purple and in red, respectively. Muropeptide analysis was performed for the BW25113 strain (black) and the $\Delta yafK$ mutant (orange). Data are the average of two biological replicates. Rel. abundance, relative abundance, defined as the ratio (%) between the intensity of the indicated isotopologue and the sum of the intensities of all four isotopologues. This implies that an isotopologue that is present at the medium switch and is neither synthesized nor degraded during the following incubation of 60 min (one generation) will display a 50% reduction in its relative abundance because the total number of stem peptides doubles in one generation.

Hydrolysis of tripeptide → Lpp bonds by YafK contributes to the shuffling of Lpp between old and new peptidoglycan stems

At the beginning of the kinetics (0 to 25 min), the absence of YafK had little impact on the relative abundance of old → new isotopologues (**Fig. 3C**). This is expected since the bulk of the PG contains old tetrapeptide stems that act as donors in the transpeptidation reaction, thus leading to the tethering of newly synthesized Lpp to old PG. At later times, the proportion of old → new isotopologues levels off earlier in the BW25113 strain than in the $\Delta yafK$ mutant. This may be accounted for by the hydrolysis of old → new isotopologues by YafK, thus generating light Lpp molecules that can be cross-linked to light tetrapeptide donor stems, the abundance of which increases after the medium switch. Thus, the larger relative abundance of new → old isotopologues in the $\Delta yafK$ mutant provides a first indication that hydrolysis of tripeptide → Lpp bonds by YafK results in the shuffling of Lpp between PG stem peptides.

The relative abundance of new → old isotopologues was higher in the BW25113 strain than in the $\Delta yafK$ mutant at all incubation times (**Fig. 3D**). The heavy Lpp molecules used for the formation of these isotopologues are expected to originate both from the old pool of free Lpp and from the YafK-mediated hydrolysis of old → old isotopologues. Thus, the reduction in the relative abundance of new → old isotopologues upon deletion of *yafK* additionally indicates that hydrolysis of tripeptide → Lpp bonds by YafK results in the shuffling of Lpp between PG stem peptides.

At least 50% of the tripeptide → Lpp bonds are hydrolyzed in one generation

The twofold decrease in the relative abundance of old → old isotopologues over the course of one generation, as observed in the absence of YafK (**Fig. 3B**), can be fully accounted for by the synthesis of new PG stem peptides and Lpp molecules. Indeed, if the old → old Tri → KR isotopologue present at the medium shift were not hydrolyzed by YafK, its *absolute amount* would remain constant over time. However, the *relative abundance* of the old → old isotopologue decreases by 50% in one generation because the total amount of the Tri → KR muropeptide doubles in one generation (as any of the bacterial constituents). Thus, there was no detectable loss of old → old isotopologues in the $\Delta yafK$ strain. Furthermore, the relative abundance of old Lpp present in both the old → old and new → old isotopologues decreased twofold in 60 min both in the BW25113 strain and in the $\Delta yafK$ mutant (**Fig. 4A** and **4B** and **Supplementary Table S2**). This implies that hydrolysis of tripeptide → Lpp bonds by YafK was not associated with any proteolytic degradation of the released Lpp. In contrast, the relative abundance of old tripeptide stems present in both the old → old and old → new isotopologues decreased more rapidly in the BW25113 strain than in the $\Delta yafK$ mutant (four *versus* twofold in 60 min; **Fig. 4C** and **4D**, respectively). Thus, hydrolysis of the Tri → KR bond by YafK resulted in a twofold decrease in the relative abundance of old tripeptide stems in these isotopologues, providing an estimate of 50% for the proportion of tripeptide → Lpp bonds hydrolyzed in one generation. This is a low estimate since old Lpp released by YafK can be cross-linked to a donor containing an old tetrapeptide stem, a reaction that regenerates old → old and old → new isotopologues.

Conclusions

Here we show that Lpp is not tethered to PG precursors prior to their polymerization, as is the case for the reaction catalyzed by sortases in Gram-positive bacteria (Dramsi et al., 2008). Instead, Lpp is incorporated into the expanding lateral cell walls independently from their sites of expansion. The free form of Lpp faces the bacterial surfaces raising the possibility that pools of free and PG-tethered Lpp might not be in dynamic equilibrium (Asmar and Collet, 2018). This is not the case because new → old isotopologues are formed in the absence of YafK (**Fig. 3**), showing that existing (old) Lpp is translocated from the outer leaflet of the OM to the periplasm and

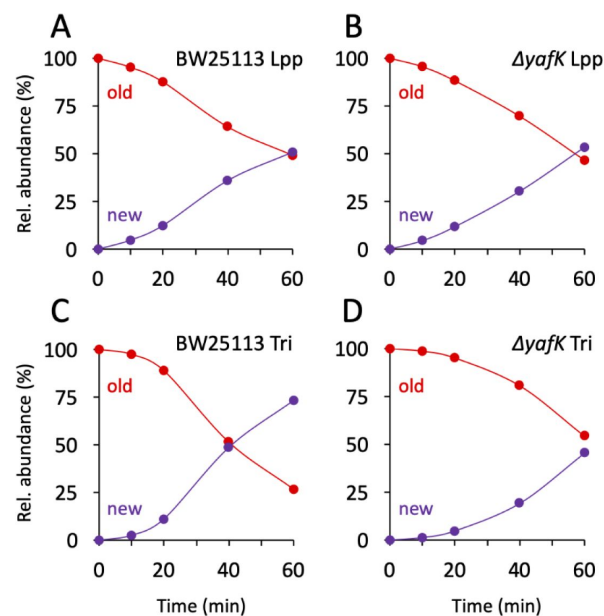


Figure 4.

Kinetics of variations in the proportions of old and new moieties in the Tri → KR mucopeptide.

(A) and (B), relative abundance of old (red) and new (purple) Lpp moieties in the isotopologues from the BW25113 strain and the $\Delta yafK$ mutant, respectively. The relative abundances were calculated from data appearing in Fig. 3 by adding the relative abundances of the new → old and old → old isotopologues (red curves; old Lpp) or of the new → new and old → new isotopologues (purple curves; new Lpp). (C) and (D), relative abundance of old (red) and new (purple) tripeptide moieties in the isotopologues from the BW25113 strain and the $\Delta yafK$ mutant, respectively. The relative abundances were calculated from data appearing in Fig. 3 by adding the relative abundances of the old → old and old → new isotopologues (red curves; old tripeptide) or of the new → new and new → old isotopologues (purple curves; new tripeptide).

tethered to neo-synthesized PG subunits. Since Lpp tethered to PG is expected to prevent OM budding, it has been proposed that vesicles are released from OM domains lacking PG-tethered Lpp (Schwechheimer et al., 2014 [↗](#)) (McMillan and Kuehn, 2021 [↗](#)). The dynamic equilibrium between free and PG-tethered Lpp may therefore be physiologically relevant to spatially control the formation of OM vesicles, and thereby, their periplasmic content.

Materials and methods

To investigate the mode of tethering of the Braun lipoprotein to PG, the *E. coli* BW25113 strain, which produces the full complement of six L,D-transpeptidases, and its derivative obtained by deletion of the *yaqK* gene, were grown in minimal medium containing [^{13}C]glucose and [^{15}N]ammonium chloride to obtain fully labeled PG (all procedures are described in details in (Atze et al., 2022)). At an optical density of 0.4 at 600 nm, bacteria were centrifuged, resuspended in a minimal medium with the same chemical composition but containing the light isotopes of carbon and nitrogen ([^{12}C] and [^{14}N]) for a continued incubation of 60 min corresponding to one generation. Culture samples were collected immediately before the medium switch (time = 0) and at four times after the medium switch in order to characterize the kinetics of the replacement of heavy (existing) by light (newly synthesized) isotopes. The PG macromolecule was extracted from these culture samples by the hot SDS procedure, digested with pronase and trypsin, and treated with muramidases, which cleave the β -glycosidic bond connecting MurNAc to GlcNAc residues. This procedure led to the release of soluble disaccharide-peptide fragments (muropeptides) that were reduced with sodium borohydride (conversion of MurNAc to muramitol residues) and purified by *rp*HPLC. The isotopic compositions of the Tri $\rightarrow$ KR isotopologues was determined by mass spectrometry and tandem mass spectrometry.

Deduced structure of fragments	Mass of ions generated by fragmentation of indicated isotopologues ^a							
	Tri→KR		Tri→KR		Tri→KR		Tri→KR	
	M _{cal}	M _{obs}	M _{cal}	M _{obs}	M _{cal}	M _{obs}	M _{cal}	M _{obs}
GlcNAc-MurNAc ^R -L-Ala-D-iGlu-DAP→L-Lys-L-Arg ^b	1,154.57	1,154.56	1,172.59	1,172.58	1,194.66	1,194.65	1,212.69	1,212.68
MurNAc ^R -L-Ala-D-iGlu-DAP→L-Lys-L-Arg	951.49	951.49	969.51	969.52	982.56	982.56	1,000.58	1,000.58
L-Ala-D-iGlu-DAP→L-Lys-L-Arg	674.37	674.38	692.39	692.39	693.41	693.41	711.43	711.43
D-iGlu-DAP→L-Lys-L-Arg	603.33	603.34	621.36	621.37	618.37	618.36	636.39	636.39
DAP→L-Lys-L-Arg	474.29	474.29	492.31	492.31	483.31	483.31	501.33	501.33
L-Lys-L-Arg	302.21	302.21	320.23	320.23	302.21	302.21	320.23	320.23
L-Arg	174.11	174.11	184.12	184.12	174.11	174.11	184.12	184.12

^aNewly synthesized and existing moieties are indicated in purple and red, respectively. M_{cal} and M_{obs}, calculated and observed monoisotopic masses (Da), respectively.

^bParental ion

Table S1.
MS-MS analysis of the four isotopologues of the Tri – KR muropeptide.

Relative abundance (%) of isotopologues at 0, 10, 20, 40, and 60 min											
Tri→KR isotopologues	Biological repeat	WT					<i>ΔyafK</i>				
		0	10	20	40	60	0	10	20	40	60
new→new	1	0	0.57	3.26	17.72	38.11	0	0.40	2.52	17.26	39.85
	2	0	1.26	2.94	21.68	40.93	0	1.09	1.49	9.04	30.84
	3	0	0.42	1.80	33.55	45.44	0	0.72	2.76	10.21	40.14
	4		0.47							13.37	
	5		0.17								
	Mean	0	0.58	2.67	24.32	41.49	0	0.74	2.26	12.17	36.94
	SD	0	0.41	0.77	8.24	3.69	0	0.34	0.68	4.45	5.29
new→old	1	0	2.88	9.99	25.14	32.34	0	0.74	2.90	7.92	9.75
	2	0	2.77	9.31	28.05	32.51	0	0.46	0.46	5.93	7.07
	3	0	1.00	6.15	19.50	30.41	0	0.72	2.76	10.21	8.91
	4		2.12							6.42	
	5		1.14								
	Mean	0	1.98	8.48	24.23	31.75	0	0.64	2.04	7.62	8.58
	SD	0	0.88	2.05	4.35	1.17	0	0.15	1.37	1.92	1.37
old→new	1	0	6.00	11.42	13.65	9.73	0	5.77	13.22	21.45	18.60
	2	0	3.97	9.21	11.36	9.60	0	2.48	7.43	16.88	15.41
	3	0	2.79	8.59	9.77	8.51	0	2.74	7.62	16.61	15.06
	4		4.48							16.11	
	5		3.56								
	Mean	0	4.16	9.74	11.59	9.28	0	3.66	9.42	17.76	16.36
	SD	0	1.20	1.49	1.95	0.67	0	1.83	3.29	2.48	1.95
old→old	1	100	90.55	75.33	43.49	19.81	100	93.09	81.36	53.37	31.80
	2	100	92.01	78.54	38.92	16.97	100	95.93	89.01	67.96	46.67
	3	100	95.79	83.45	37.19	15.64	100	95.93	86.77	66.31	35.89
	4		92.93							64.10	
	5		95.13								
	Mean	100	93.28	79.11	39.86	17.47	100	94.98	85.71	62.94	38.12
	SD	0	2.18	4.09	3.25	2.13	0	1.64	3.93	6.57	7.68

Supplementary Table S1.

Kinetic analysis of the relative abundance of Tri→KR isotopologues in wild-type and *ΔyafK*.

Relative abundance (%) at 0, 10, 20, 40, and 60 min											
Combinations of Tri→KR isotopologues	Biological repeat	WT					ΔyafK				
		0	10	20	40	60	0	10	20	40	60
Containing light KR new→new + old→new	1	0	6.57	14.68	31.38	47.85	0	6.17	15.74	38.71	58.45
	2	0	5.23	12.15	33.03	50.53	0	3.57	8.92	25.92	46.25
	3	0	3.21	10.39	43.32	53.95	0	3.46	10.38	26.81	55.20
	4		4.95							29.48	
	5		3.73								
	Mean	0	4.74	12.41	35.91	50.77	0	4.40	11.68	29.93	53.30
	SD	0	1.61	2.25	10.19	4.36	0	2.17	3.97	6.93	7.23
Containing heavy KR new→old + old→old	1	100	93.43	85.32	68.62	52.15	100	93.83	84.26	61.29	41.55
	2	100	94.77	87.85	66.97	49.47	100	96.39	89.47	73.89	53.75
	3	100	96.79	89.61	56.68	46.05	100	96.65	89.53	76.52	44.80
	4		95.05							70.52	
	5		96.27								
	Mean	100	95.26	87.59	64.09	49.23	100	95.62	87.75	70.56	46.70
	SD	0	3.06	6.14	7.60	3.30	0	1.80	5.30	8.49	9.05
Containing light Tri new→new + new→old	1	0	3.45	13.25	42.86	70.45	0	1.14	5.42	25.18	49.60
	2	0	4.03	12.25	49.73	73.44	0	1.55	1.95	14.97	37.91
	3	0	1.42	7.95	53.05	75.84	0	1.44	5.52	20.42	49.05
	4		2.60							19.79	
	5		1.31								
	Mean	0	2.56	11.15	48.54	73.25	0	1.38	4.30	19.79	45.52
	SD	0	1.29	2.82	12.59	4.86	0	0.50	2.04	6.37	6.66
Containing heavy Tri old→new + old→old	1	100	96.55	86.75	57.14	29.55	100	98.86	94.58	74.82	50.40
	2	100	95.97	87.75	50.27	26.56	100	98.41	96.44	84.84	62.09
	3	100	98.58	92.05	46.95	24.16	100	98.67	94.39	82.92	50.95
	4		97.40							80.21	
	5		98.69								
	Mean	100	97.44	88.85	51.46	26.75	100	98.65	95.13	80.70	54.48
	SD	0	3.38	5.58	5.21	2.80	0	3.47	7.22	9.05	9.63

Supplementary Table S2.

Kinetic analysis of the content of Tri→KR isotopologues in new and old Tri and KR moieties.

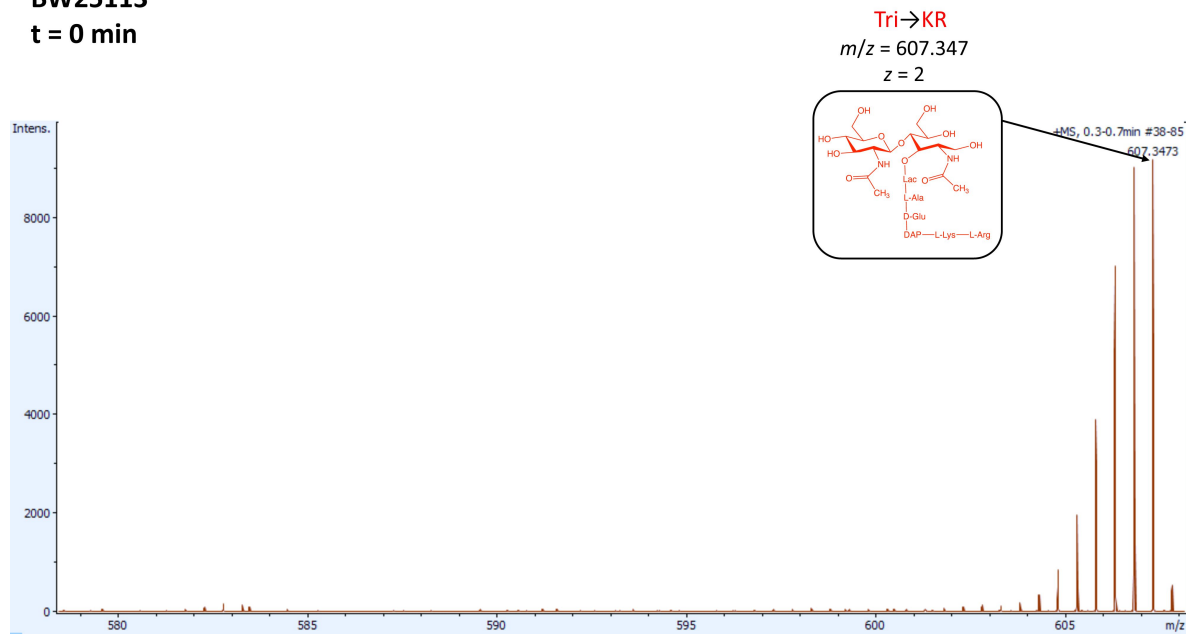
Supplementary File 1.

Time-resolved appearance of differentially labeled Tri → KR mucopeptide isotopologues containing new (light) and old (heavy) Lpp and PG moieties.

This file contains mass spectral data obtained for the kinetics analyses for the BW25113 strain and the *ΔyafK* mutant.

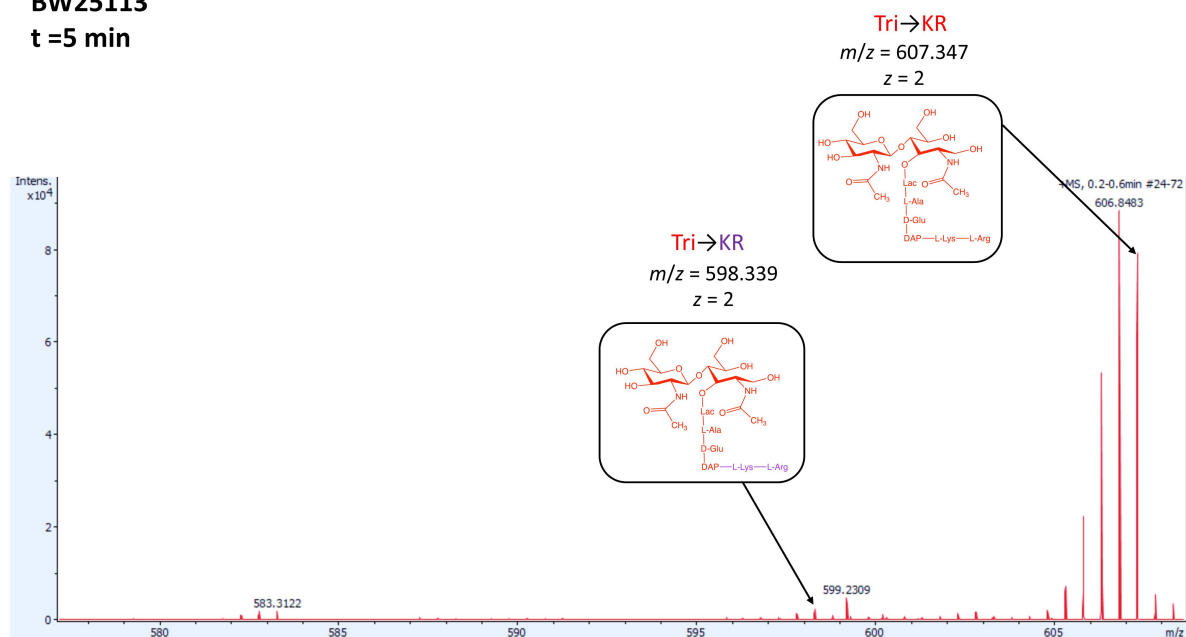
BW25113

t = 0 min



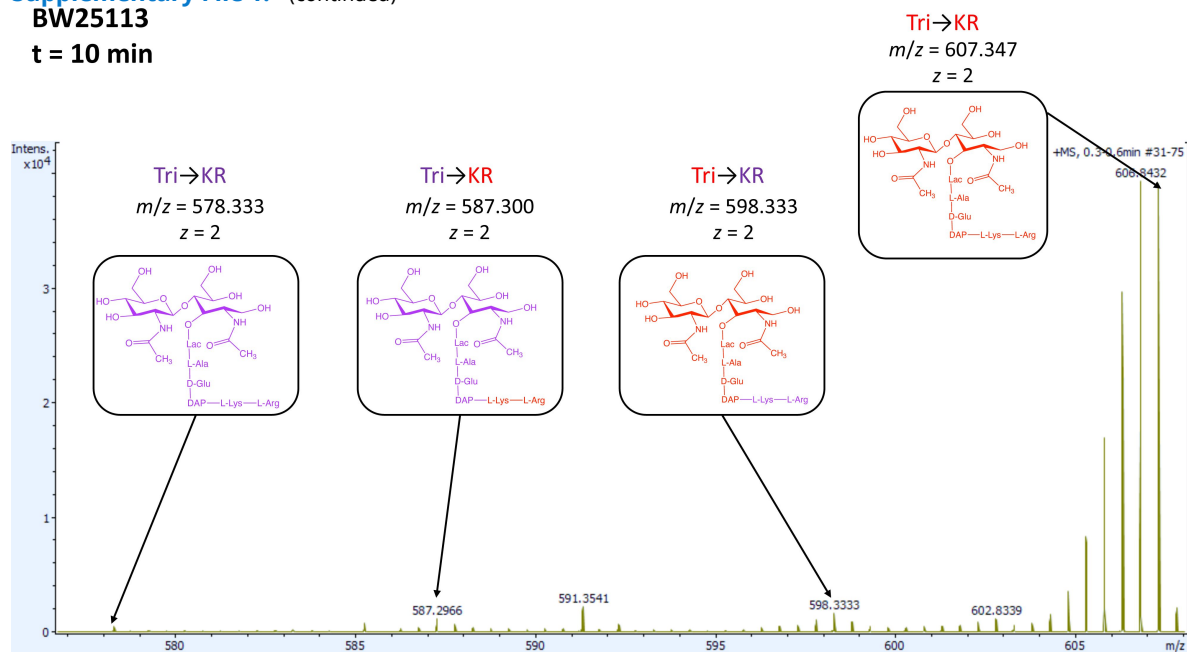
BW25113

t = 5 min



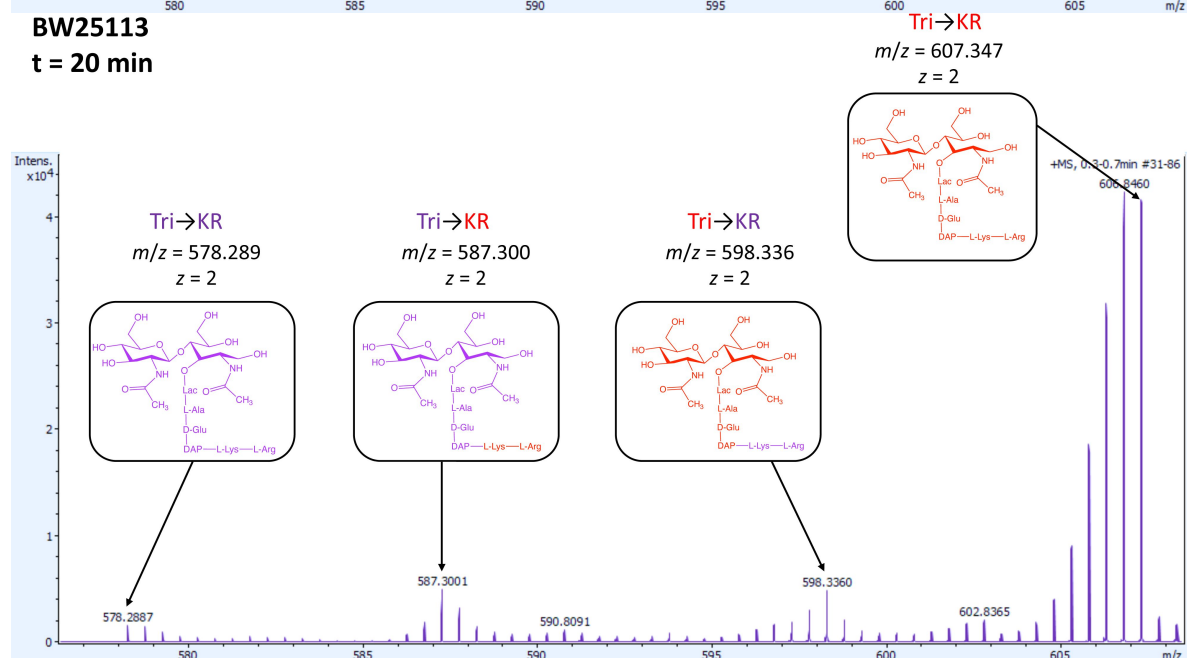
Supplementary File 1. (continued)
BW25113

t = 10 min



BW25113

t = 20 min



Supplementary File 1. (continued)

BW25113 Tri→KR

t = 40 min $m/z = 578.289$
 $z = 2$

Tri→KR

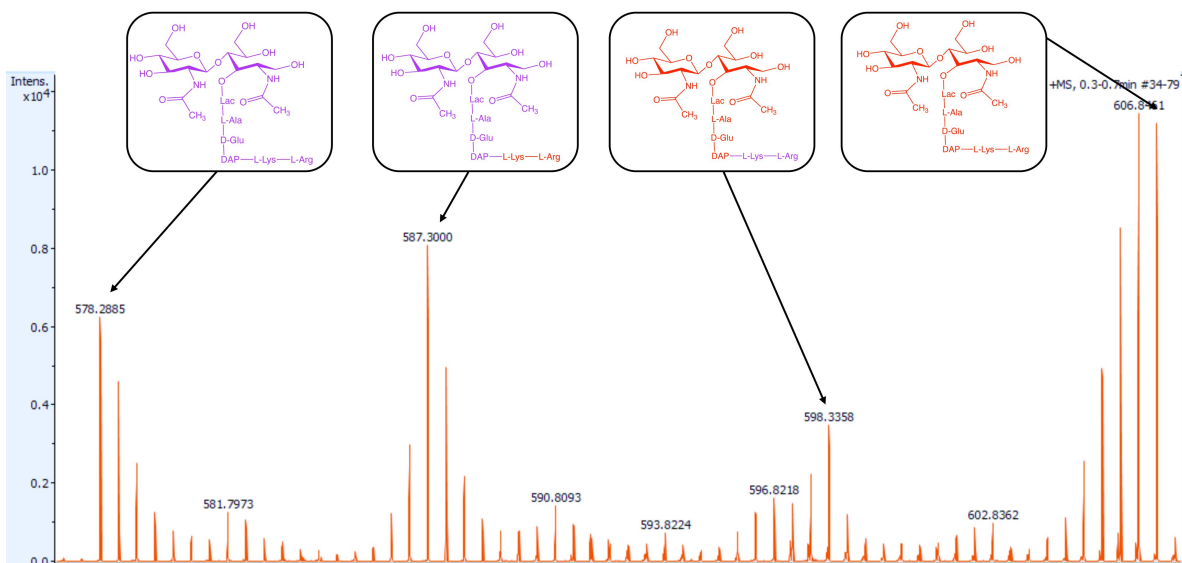
$m/z = 587.300$
 $z = 2$

Tri→KR

$m/z = 598.336$
 $z = 2$

Tri→KR

$m/z = 607.347$
 $z = 2$



BW25113

t = 60 min

Tri→KR

$m/z = 578.289$
 $z = 2$

Tri→KR

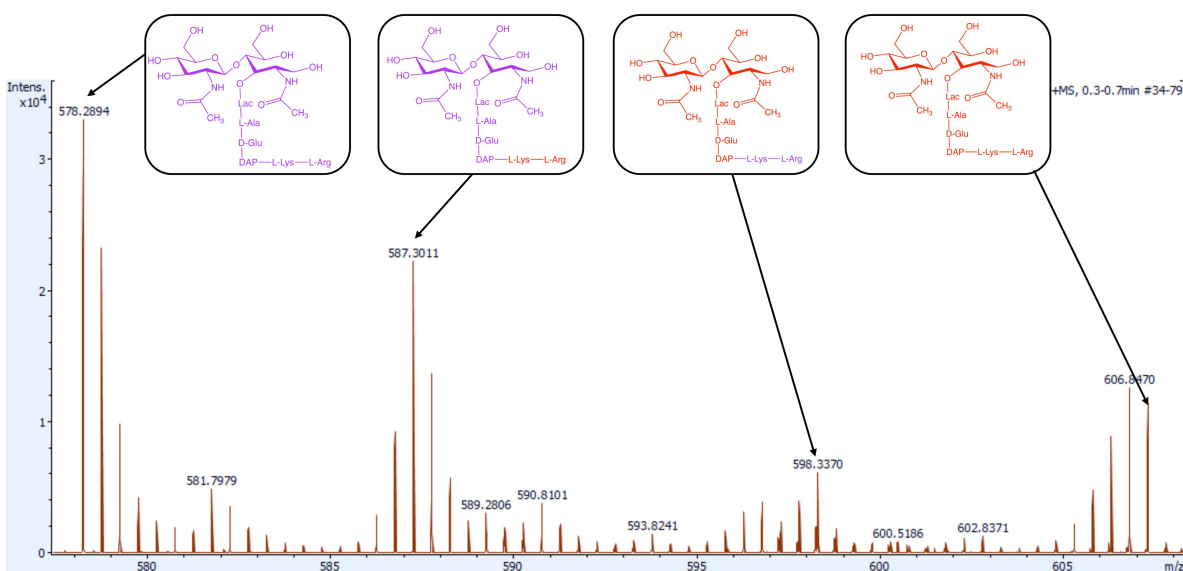
$m/z = 587.301$
 $z = 2$

Tri→KR

$m/z = 598.337$
 $z = 2$

Tri→KR

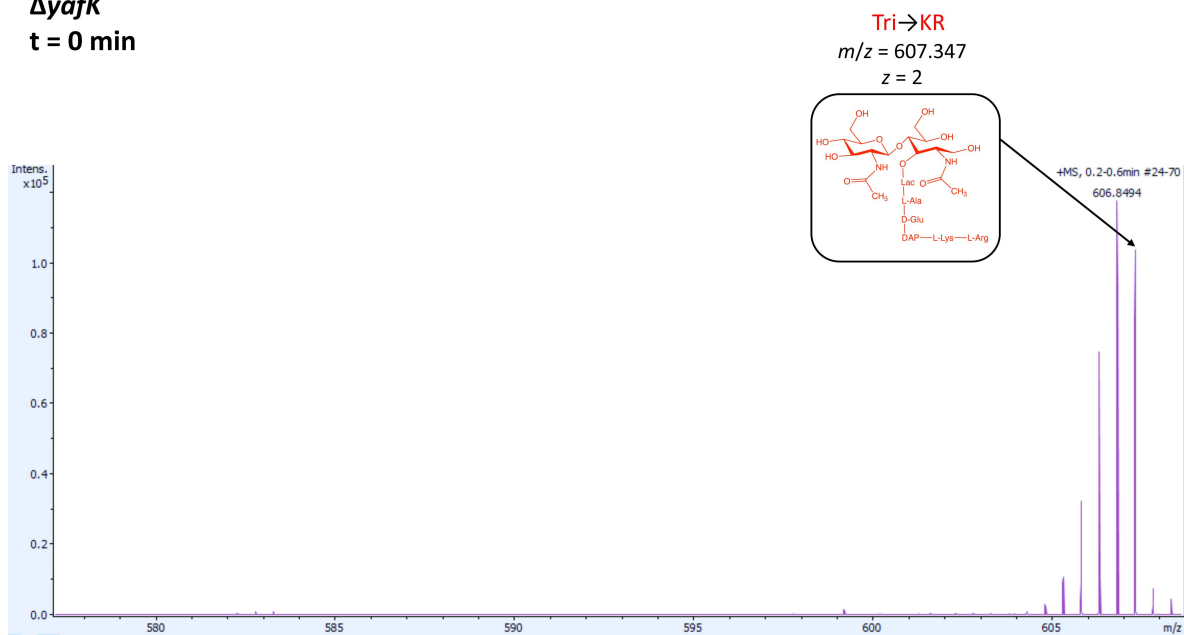
$m/z = 607.347$
 $z = 2$



Supplementary File 1. (continued)

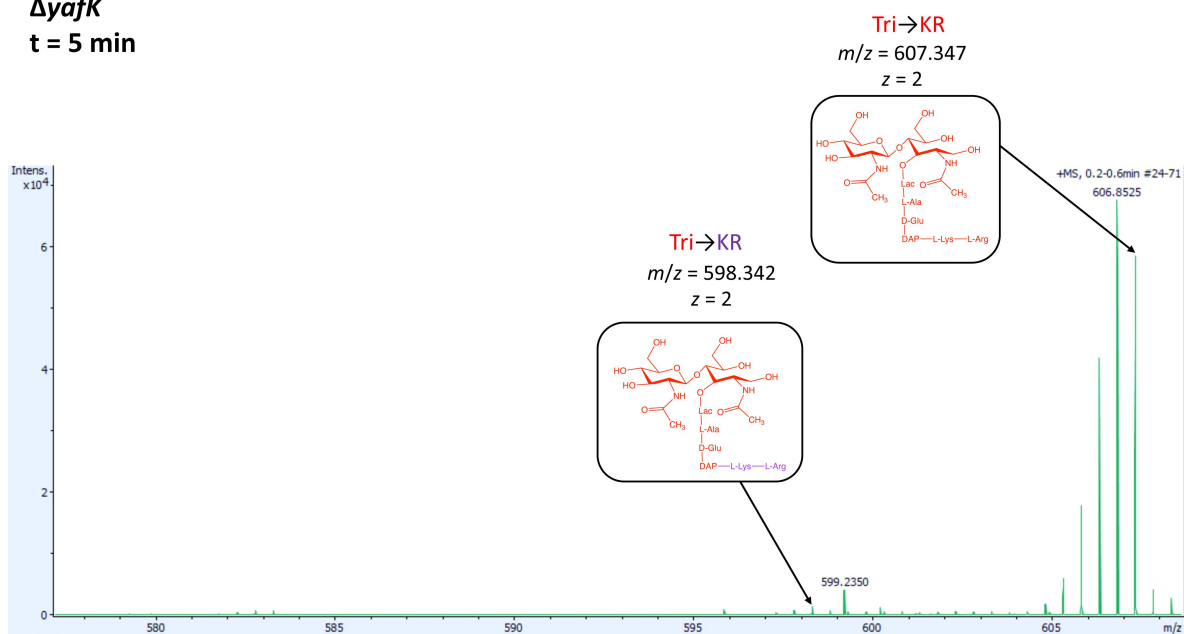
ΔyafK

t = 0 min



ΔyafK

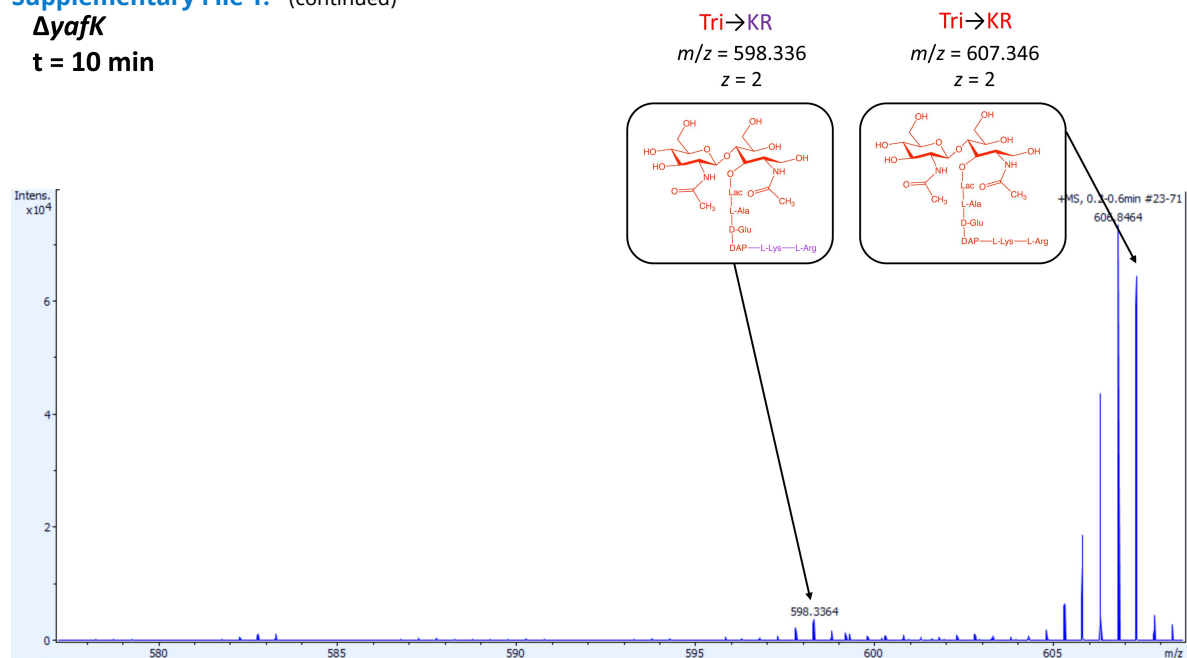
t = 5 min



Supplementary File 1. (continued)

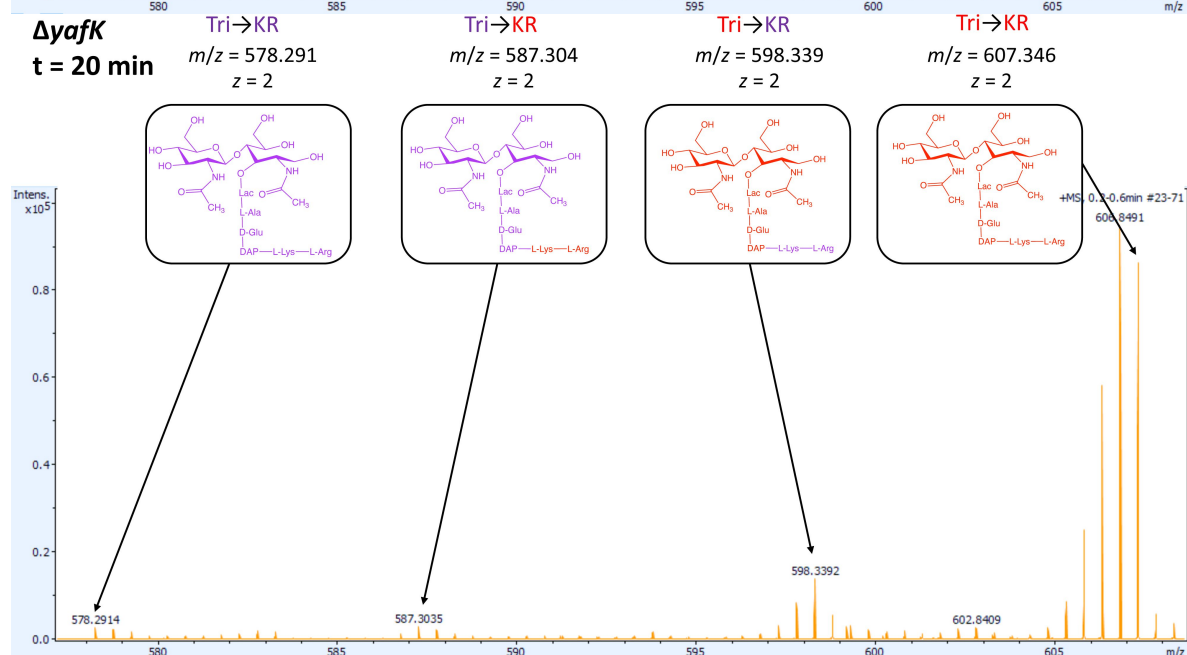
ΔyafK

t = 10 min

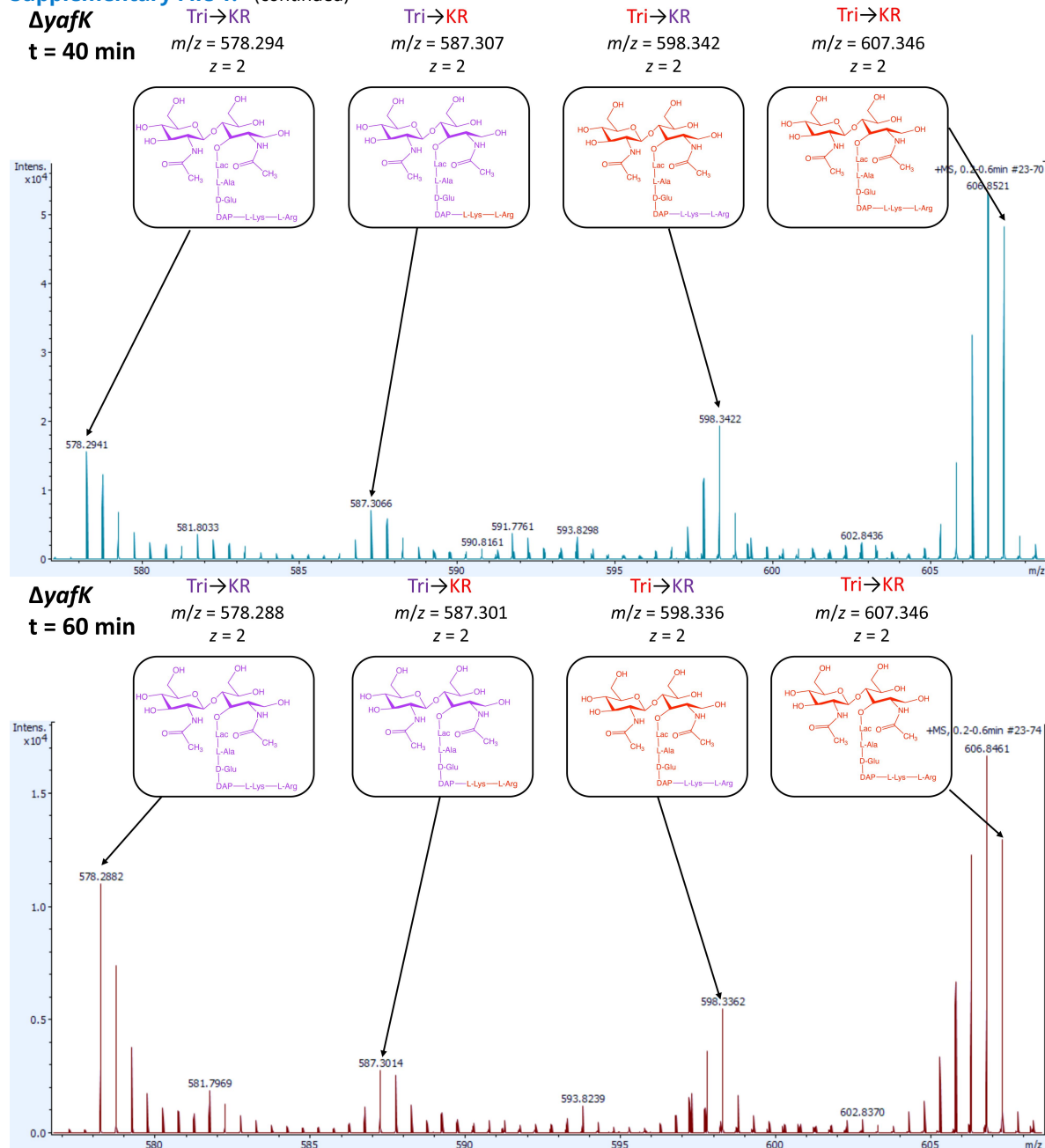


ΔyafK

t = 20 min



Supplementary File 1. (continued)



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Editors

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University of the Witwatersrand, Johannesburg, South Africa

Senior Editor

Bavesh Kana

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Reviewer #1 (Public Review):

The authors present data on outer membrane vesicle (OMV) production in different mutants, but they state that this is beyond the scope of the current manuscript, which I disagree with. This data could provide valuable physiological context that is otherwise lacking. The preliminary blots suggest that YafK does not alter OMV biogenesis. I recommend repeating these blots with appropriate controls, such as blotting for proteins in the culture media, an IM protein, periplasmic protein and an OM protein to strengthen the reliability of these findings. Including this data in the manuscript, even if it does not directly support the initial hypothesis, would enhance the physiological relevance of the study. Currently, the manuscript relies completely on the experimental setup (labeling-mass spec) previously developed by the authors, which limits the broader scope and interpretability of this study.

Additionally susceptibility of strains to detergents like SDS can be tested to provide a much needed physiological context to the study.

In summary, the authors should consider revising the manuscript to improve clarity, substantiate their claims with more detailed evidence, and include additional experimental results that provide necessary physiological context to their study.

<https://doi.org/10.7554/eLife.91598.2.sa2>

Reviewer #2 (Public Review):

Summary:

The authors of this study have sought to better understand the timing and location of the attachment of the lpp lipoprotein to the peptidoglycan in *E. coli*, and to determine whether YafK is the hydrolase that cleaves lpp from the peptidoglycan.

Strengths:

The method is relatively straightforward. The authors are able to draw some clear conclusions from their results, that lpp molecules get cleaved from the peptidoglycan and then re-attached, and that YafK is important for that cleavage.

Weaknesses:

Figure 3 and 4 - why are the data shown here only two biological replicates, when there are 3-5 replicates shown in table S1 and S2? This makes it seem like you are cherry picking your favorite replicates. Please present the data as the mean of all the replicates performed, with error shown on the graph.

This work will have a moderate impact on the field of research in which the connections between the OM and peptidoglycan are being studied in *E. coli*. Since lpp is not widely conserved in gram negatives, the impact across species is not clear. The authors do not discuss the impact of their work in depth.

<https://doi.org/10.7554/eLife.91598.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Weaknesses:

- Only one mutant (YafK) is used to make the conclusion.

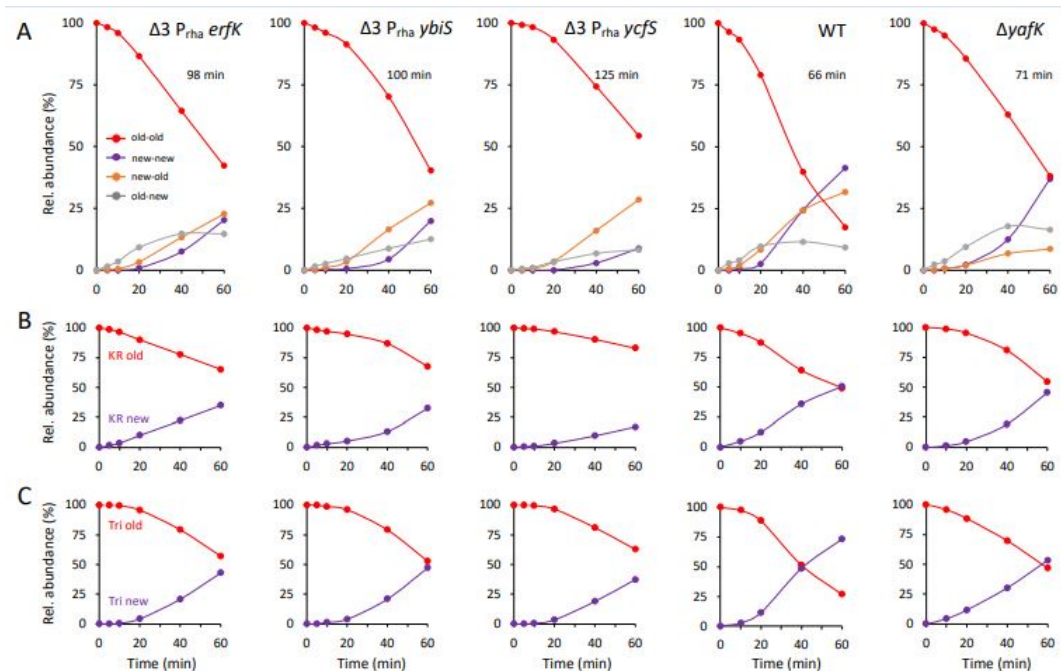
The aim of the study is to determine the effect of the hydrolysis of the PG → Lpp bond on the dynamics of the tethering of Lpp to PG. Since YafK is the only enzyme catalyzing this reaction, it is appropriate to compare the wild-type strain to an isogenic yafK deletion mutant. Nonetheless, we carefully consider this comment and will investigate the dynamics of the tethering of Lpp to PG in mutants deficient in the production of the L,D-transpeptidases responsible for tethering Lpp to PG.

Additional kinetic analyses were performed on strains relying on a single L,D-transpeptidase for LPP tethering to PG. *Escherichia coli* produces three L,D-transpeptidases catalyzing the tethering of LPP to PG (Ybis, YcfS, and ErfK). The corresponding genes were deleted from the chromosome of strain BW25113, thus generating strain BW25113Δ3. Plasmids encoding each one of these three enzymes were independently introduced in BW25113Δ3. Qualitatively, LC-MS analyses revealed similar kinetics for the four Tri-KR isotopologues purified from wild-type strain BW25113 and from the three BW25113Δ3 derivatives producing a single plasmidencoded L,D-transpeptidase (Ybis, YcfS, or ErfK) under the control of a rhamnose inducible promoter (Prha) of plasmid pHV30 (Voedts et al. EMBO J. 2021 40:e108126, doi: 10.15252/emj.2021108126) (see panel A in figure 1 below). Briefly, and as indicated in the first version of the main text, the old → new Tri → KR isotopologue was first synthesized. The new → new isotopologue was not detected 5 min after the medium switch. These results

indicate that the newly-synthesized PG disaccharidepeptide subunits and Lpp are independently incorporated into the expanding PG polymer. The proportion of the new → old isotopologue exceeded that of the old → new isotopologue at around 40 min (for the strain producing ErfK) or 20 min (for the strains producing YbiS or YcfS). This is the hallmark of the activity of the YafK hydrolase that liberates existing (old) Lpp that can be tethered to newly synthesized disaccharide-peptide subunit thereby generating the new → old isotopologue. In absence of the YafK hydrolase, the relative proportion of the new → old isotopologue is lower since this isotopologue can only result from the tethering of the preexisting free forms of Lpp to newly synthesized disaccharide-peptide units. The contribution of YafK to variations in the relative abundance of the four isotopologues was also investigated by combining the relative abundance of isotopologues containing either old versus new KR (panel B) or old versus new PG stem peptide (panel C) moieties. As discussed in the first version of the manuscript for strains BW25113 and BW25113ΔyafK, this analysis revealed that the existing (old) disaccharide-tripeptide moieties in the Tri → RK isotopologues disappears more rapidly than the existing (old) KR moieties due to the hydrolysis of the old → old Tri-KR isotopologue by YafK. These results indicate that the mode of tethering of Lpp to PG and the dynamic equilibrium between the PG-tethered and free forms of Lpp are similar for the YbiS, YcfS, and ErfK L,D-transpeptidases. Quantitatively, we also noticed that the overall decrease in the relative abundance of all Tri → KR isotopologues containing existing (old) moieties was slower for the strains producing only ErfK, YbiS, or YcfS than for the wild type and ΔyafK strains. This could be accounted for by an increase in the generation time of the former group of three strains. This is a limitation of our study because it precludes the comparison of the evolution of a particular isotopologue in several strains, as performed in Fig. 3 for strains BW25113 and BW25113ΔyafK. For this reason, we prefer to present these data in the rebuttal rather than in the manuscript. Indeed, presentation of the data in the main text would require introducing a new mode of presentation of the data (variations in the relative abundance of all four isotopologues in the same strain; see figure below) in addition to variations of the relative abundance of any one of the four isotopologues between strains (Fig. 3). Introduction of this additional mode of presentation of the data would complicate the initial manuscript in an unnecessary manner because the data obtained with mutants producing a single L,D-transpeptidase (ErfK, YbiS, or YcfS) confirmed the data obtained with the wild-type strains producing the three L,D-transpeptidases.

Author response image 1.

MS-based kinetic analysis of Lpp tethering to PG.



-Time points to analyse Tri-KR isotopologues in Wt (0,10,20,40,60 min) and yafK mutant (0,15, 25, 40, 60 min) are not the same.

The purpose of the experiments is to compare the kinetics of formation and hydrolysis of the PG → Lpp bond in the WT versus $\Delta yafK$ strains. Comparison of the kinetics is therefore possible even though the kinetics are not based on the exact same time points. Nonetheless, we will reproduce the kinetics experiment (see also answers to Reviewer 2) and use the same time points in these additional experiments.

We have performed additional analyses to provide kinetic data for at least three biological repeats and for the same periods of incubation after the medium switch (0, 10, 20, 40, and 60 min). The full set of data, including means and standard deviations, appear in the additional Table S1. We have also updated Fig. 3 with the means calculated with these additional values. The conclusions of the first version of the manuscript are fully supported by the additional data requested by the reviewer. We have also revised Fig. 4 based on the full set of data appearing in Table S2.

Reviewer #2 (Public Review):

Weaknesses:

- However, the authors make a few other conclusions from their data which are harder to understand the logic of, or to feel confident in based on the existing data. They claim that their 5-time point kinetic data indicates that new lpp is not substantially added to lipidII before it is added to the peptidoglycan, and that instead lpp is attached primarily to old peptidoglycan. I believe that this conclusion comes from the comparison of Fig.s 3A and 3C, where it appears that new lpp is added to old peptidoglycan a few minutes before new lpp is added to new peptidoglycan. However, the very small difference in the timing of this result, the minimal number of time points and the complete lack of any presentation of calculated error in any of the data make this conclusion very tenuous. In addition, the authors conclude that lpp is not significantly attached to septal

peptidoglycan. The logic behind this conclusion appears to be based on the same data, but the authors do not provide a quantitative model to support this idea.

The reviewer is correct in stating that we claim that Lpp is not substantially added to lipid II before incorporation of the disaccharide-pentapeptide subunit into the expanding PG network. This conclusion is based on the paucity of PG-Lpp covalent adducts containing light PG and Lpp moieties at the earliest time points. To substantiate more thoroughly this finding, we will reproduce the kinetic experiments with more early time points. The paucity of the new → new PG-Lpp isotopologues also implies that Lpp might not be extensively tethered to septal peptidoglycan since the latter is assembled from newly synthesized PG (see our previous publication Atze et al. 2021 and references therein). Quantitatively, septal synthesis roughly accounts for one third of the total PG synthesis. It is therefore expected that tethering of Lpp to septal PG would represent one third of the total number of newly synthesized Lpp molecules tethered to PG. We therefore proposed that the paucity of new → new PG-Lpp isotopologues at early time points of the kinetics implies that Lpp is preferentially tethered to the side wall. This is only one of several conclusions that we reach in the present study and we were very careful in the wording of our results.

We would first like to stress that our claim that Lpp is primarily attached to old peptidoglycan rather than to lipid II is indeed supported by the results presented in the first version of the manuscript. In fact, the opposite mechanism, i.e. Lpp linking to Lipid II, as established for the linking of proteins to PG by sortases in Gram-positive bacteria, would result in the exclusive tethering of newly synthesized Lpp to newly synthesized PG stems (Fig. 3). This is clearly not the case since the new → new isotopologues are present in small amounts 10 min after the medium switch and are not detectable at 5 min (data appearing in Table S1 and new mass spectra added to Supplementary file 1). Instead, our data indicate that newly synthesized Lpp is tethered to existing PG. Thus, the relevant comparison is not the absolute value of the delay in the appearance of isotopologues in Figs 3A and 3C, as suggested by the reviewer. Rather, the relevant comparison should take into consideration these two following modes of Lpp tethering to PG: (i) tethering Lpp to Lipid II versus (ii) tethering of Lpp to existing PG independently from insertion of new subunits into the expanding PG. The former mode implies the exclusive formation of new → new isotopologues, which were not detected at early time points. The latter mode implies the prevalent formation of old → new isotopologues that were indeed preponderant at early time-points. Thus, our analysis clearly eliminates the first mode of Lpp tethering to PG (tethering of Lpp to Lipid II) and validates the second one (tethering of Lpp to existing PG). As stated in our answers to reviewer 1, we have generated additional repeats and the full set of data, including means and SD values, appears in the additional Supplementary Tables S1 and S2.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

-All major reactions catalysed by L,D-transpeptidases must be studied using the labeling-mass spec technique and compared with YafK to strengthen the conclusions.

As described above (Figure 1), we explored the dynamics of Lpp tethering in mutants producing a single L,D-transpeptidase.

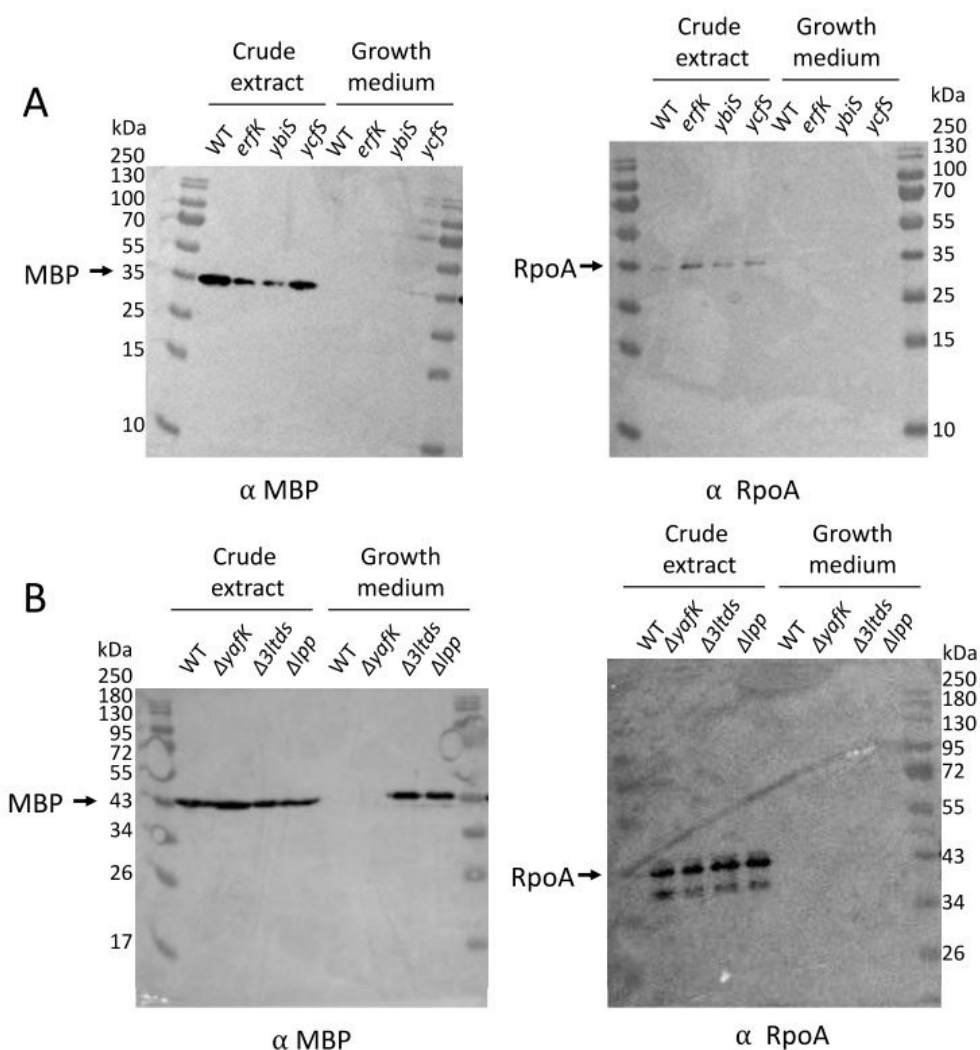
-Experiments on the effect of YafK on the bacterial envelope and production of vesicles should be concluded to support the claims.

We have analyzed the extent of outer membrane vesicle (OMV) formation both in the wild type strain and in each one of the mutant strains characterized in this study by using a procedure described in detail in one of our previous publications (Hugonneau-Beaufet et al.

Microbiol Spectr. 2023 11:e0521722, doi: 10.1128/spectrum.05217-22). Figure 2 below shows that loss of Lpp or of its tethering to PG, following deletion of genes encoding L,D-transpeptidases ErfK, YbiS, and YcfS, results in the formation of OMVs as revealed by the presence of the maltose-binding protein (MBP, 42 kDa) in the corresponding spare culture medium (as detected by immunoblotting). The RNA polymerase subunit RpoA (36 kDa), used as a control, was not detected in these spare culture media, indicating that loss of either Lpp alone or of ErfK, YbiS, and YcfS together was not associated with bacterial lysis. This analysis also showed that production of ErfK, YbiS, or YcfS alone was sufficient to prevent formation of OMVs. Finally, deletion of YafK, as expected, did not lead to OMV formation. These confirmatory results are out of the scope of the manuscript that focuses on the dynamics of Lpp tethering to PG rather than on the role of that tethering in the envelope stability.

Author response image 2.

Figure 2. Immuno-detection of OMV formation.



Reviewer #2 (Recommendations For The Authors):

- Why so much background about previous results in the abstract? Previous results don't seem required for understanding the description of new results here. Maybe put a

sentence about importance at the end, instead.

The background information is important for two reasons. First, because it is important to stress that the method used to determine the structure and dynamics of the isotopologues is novel and has been validated in various ways, including the modeling of isotopic clusters, in a previous study (<https://doi.org/10.7554/eLife.72863>). Since the current study is an extension of this previous report it is relevant to introduce the type of information that can be obtained by this approach. Second, because it is also important to stress that kinetic analyses have been previously reported for the incorporation of disaccharide-peptide units into the expanding peptidoglycan (<https://doi.org/10.7554/eLife.72863>). In the current study, we focused on the mode of Lpp-to-PG tethering in the context of PG expansion that thus had to be introduced.

- Abstract: tethering of lpp to septal pg is limited by what? Limited to what? Wording not clear.

The unclear sentence has been rephrased. Revised version “Newly synthesized septum PG appears to contain small amounts of tethered Lpp.”

- The figure legend for fig 1b - I only see one red double arrow?

Black double arrows indicate the position of glycosidic bonds cleaved by the muramidases. Their size was increased so that they appear more distinctly in the image.

- Fig 3 and Fig 4- these should be shown with error.

The full set of data with means and standard deviations appear in Supplementary Tables S1 and S2.

- This new-> old, old-> new annotation is confusing. Is the PG fragment or the lpp old or new? Are you distinguishing between which part is old and new by the ordering? Or, could either the PG fragment or the lpp be old to be annotated as old-> new? I think you are trying to explain it in the figure 3CD legend, but it could be presented more clearly. When you say respectively, do you mean that old->new means old muropeptide, new lpp? And new-> old means new muropeptide and old lpp? Why not just use the same annotation system you use in fig 2? Or, use subscripts to indicate old and new?.

The designation of isotopologues is correct and adequate to designate the products of transpeptidation catalyzed both by PBPs and L,D-transpeptidases. This nomenclature of transpeptidation products has been introduced in the 70s (see Schleifer and Kandler 1972 Bacteriological Reviews 36:407-477). In this bond designation, the acyl donor and the acyl acceptor appear left and right, respectively, separated by an arrow to indicate the CO-to-NH polarity of the amide bond. For the Tri → KR isotopologues, the peptide stem acts as the acyl donor whereas Lpp acts as the acyl acceptor. There is therefore no ambiguity in the annotation. This also applies to the old → new-type annotation, old (existing) PG stem linked to new (neosynthesized) Lpp. In the figures, we used a color code to identify old (red) and new (purple) in the Tri → KR moieties. Since a color code cannot be used in the main text, we used the old → new-type of annotation. A sentence has been added at the end of the legend to Fig. 1b to introduce this nomenclature “Please note that we used the standard nomenclature for transpeptidation products in which the acyl donor and the acyl acceptor appear left and right, respectively, separated by an arrow to indicate the CO-to-NH polarity of the amide bond”.

- Pg 5 - first paragraph. I'm struggling with the logic of your conclusion that lpp is not attached to lipid II - it seems that this conclusion is based on the timing of the appearance of the hybrid isotopes. You say you would expect the new-new ones to appear quickly, but how quickly would you expect that, and why? You do see new-new ones appearing fairly quickly, in 20 minutes, so I don't understand the logic of why that timing excludes the lipidII modification model. Please elaborate further.

See answer above to reviewer 2 and analysis of samples collected shortly after the medium switch (Table S1). See also the revised version of Supplementary file 1 that shows mass spectra for peptidoglycan extracted 5 min after the medium switch.

- The conclusion about tethering of lpp to septal PG also appears to be somewhat tenuous, which the authors concede when then use the word "might" in the section of the results. However, the language in the abstract is more definitive. Please tone down the language in the abstract, or provide more evidence to support this conclusion. At the least, you could add a little discussion of the numbers. At a given time in mixed culture, how much PG is being constructed at the septum? How does that percentage line up with the rate of PG label loss vs the rate of lpp label loss?

- Pg 5, bottom paragraph. I don't know what you mean by "there was no loss of old->old in the Δ yafK strains, " when you just a sentence above described the decrease.

The data of the MS analyses are presented as the relative abundance of isotopologues. If the old $\rightarrow$ old Tri $\rightarrow$ KR isotopologue present at the medium shift were not hydrolyzed by YafK, its absolute amount would remain constant over time. However, the relative abundance of the old $\rightarrow$ old isotopologue decreases by 50% in one generation because the total amount of the Tri $\rightarrow$ KR muropeptide doubles in one generation (as any of the bacterial constituents). In Fig. 3B, we indeed observed that the relative amount of old $\rightarrow$ old isotopologue is about 50% after one generation in the Δ yafK mutant indicating the persistence of the isotopologue. In contrast, production of YafK in the strain BW25113 results in lower abundance of this isotopologue (in the order of 90%).

To better explicit the concept we expanded the reasoning in the relevant paragraph of the revised version of the manuscript.

- Pg 6 - I don't understand how you are drawing a conclusion about the proteolytic degradation of lpp from these data. Please clarify your reasoning.

In the analysis presented in Fig. 4, we investigated the relative abundance of old and new Lpp based on the relative abundance of old and new KR moieties in all four Tri-KR isotopologues. As stated in the preceding answer, the relative abundance of KR moieties should be 50% after one generation if no degradation of Lpp occurs. This is observed both for BW25113 (Fig. 4A) and for the Δ yafK mutant (Fig. 4B), thus supporting our claim that Lpp is not degraded. In contrast, the relative abundance of the old Tri moiety is lower than 50% for the wild type strain (Fig. 4C) but not for the Δ yafK mutant (Fig. 4D). This reflects the fact that YafK hydrolyzes the PG-Lpp bond and that Lpp released by this reaction can be cross-linked to neo-synthesized PG stems. Please note that, in this reaction, the substrate is a tetrapeptide donor stem (Fig. 1C).

<https://doi.org/10.7554/eLife.91598.2.sa0>